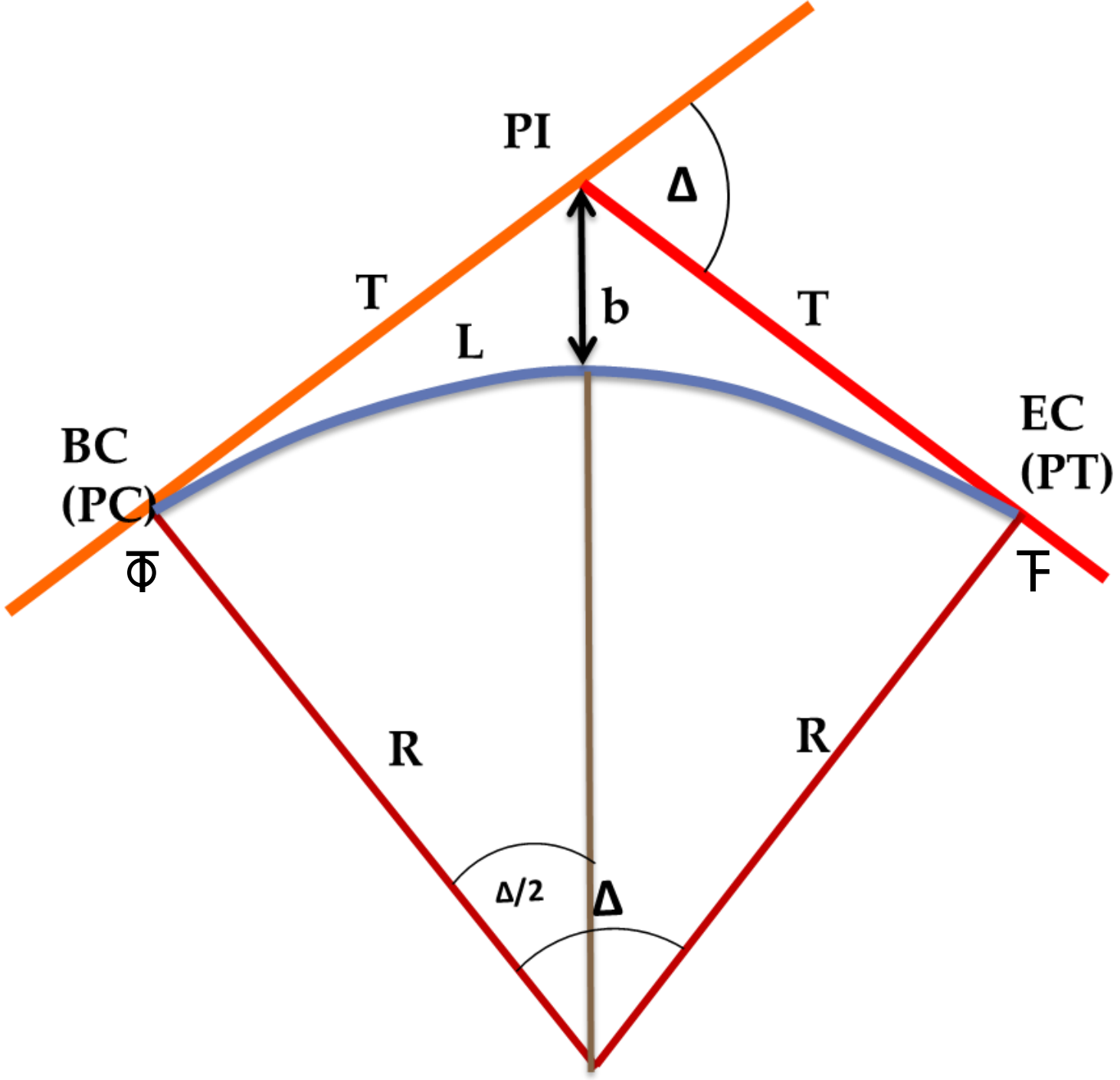
Highway Engineering

- INS3441 -

PROBLEM SESSION 3

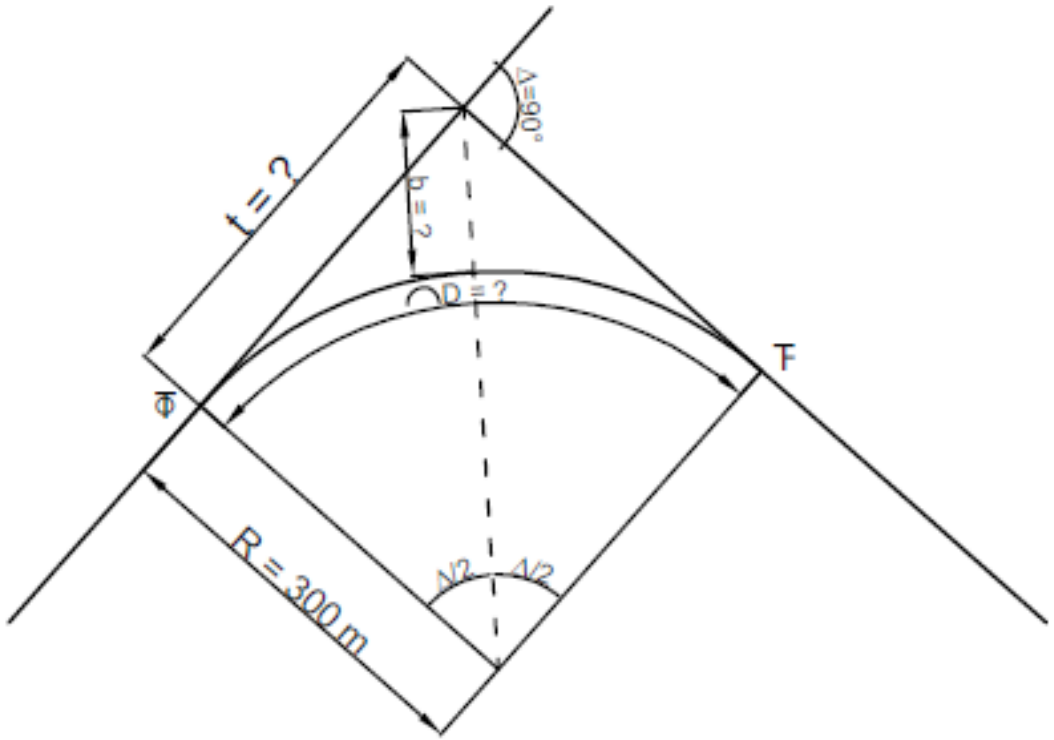


Δ = Deflection Angle
R = Radius of Circular Curve
D = Length of Curvature
t = Tangent Length
b = External Distance
BC = Beginning of Curve (PC)
EC = End of Curve (PT)
PI = Point of Intersection



Question – 1

Find all components of a horizontal curve with a radius of 300 m and curve (deflection) angle of 90º.



Solution:

$R = 300m \qquad \Delta = 90^0 \qquad t = ? \; ; \; D = ? \; ; \; b = ?$

$$t = R * tg(\frac{\Delta}{2})$$

$$t = 300 * tg(\frac{90}{2}) = 300\,m$$

$$D = 2 * \pi * R * \frac{\Delta}{360}$$

$$D = 2 * \pi * 300 * \frac{(90)}{360} = 471{,}24\,m$$

$$b = R * \left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1 \right) = 300 * \left(\frac{1}{\cos\left(\frac{90}{2}\right)} - 1 \right) = 124{,}26m$$

Question – 2

Data of an horizontal curve is given below. Find missing components of the horizontal curve and find km values for other critical points of the horizontal curve.

Supplementary angle: 150°; Tangent length: 280.007 m; Φ_{km} *= 1+550*

Solution:

Curve angle $\Delta = 180 - 150 = 30°$

$$t = R * tg\left(\frac{\Delta}{2}\right)$$

$$280{,}007 = R * tg\left(\frac{30}{2}\right) \rightarrow R = 1045\ m$$

$$D = 2 * \pi * R * \frac{\Delta}{360}$$

$$D = 2 * \pi * 1045 * \frac{30}{360} = 547.16\ m$$

$$b = R * \left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1\right) = 1045 * \left(\frac{1}{\cos\left(\frac{30}{2}\right)} - 1\right) = 36.86\ m$$

$$F_{km} = \Phi_{km} + D = [1 + 550] + 547.16 = 2 + 097^{16}$$

$$B_{km} = \Phi_{km} + \frac{D}{2} = [1 + 550] + \frac{547.16}{2} = 1 + 823^{58}$$

Question – 3

Data about two inverse horizontal curves designed on a two-way double lane highway.

a) Find all components of the horizontal curves.
b) Calculate the distance between the inverse horizontal curves and check if the distance is enough.

Δ_1: 59.73° ; Δ_2: 68.76° ; $\Phi_{1,km}$ = 1+000 ; $\Gamma_{1,km}$ = 1+344 ; $\Phi_{2,km}$ = 1+500 ; R_2 = 250 m

Solution:

a) Components for Curve 1:

$$D_1 = \Gamma_{1,km} - \Phi_{1,km} = [1+344] - [1+000] = 344 \text{ m}$$

$$D = 2 * \pi * R * \frac{\Delta}{360} \rightarrow 344 = 2 * \pi * R_1 * \frac{59.73}{360}$$

$$\rightarrow R_1 = 330 \text{ m}$$

$$t = R * tg(\frac{\Delta}{2}) \rightarrow t_1 = 330 * tan\frac{59.73}{2} = 189.49\ m$$

$$b = R * \left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1\right)$$

$$\rightarrow b_1 = 330 * \left(\frac{1}{\cos\left(\frac{59.73}{2}\right)} - 1\right) = 50.53\ m$$

Components for Curve 2: Δ_2: 68.76° ; $\Phi_{2,km}$ = 1+500 ; R_2 = 250 m

$$D_2 = 2 * \pi * R_2 * \frac{\Delta_2}{360} = 2 * \pi * 250 * \frac{68.76}{360} = 299.87\ m$$

$$t_2 = 250 * tan\frac{68.76}{2} = 171.05\ m$$

$$b_2 = 250 * [\sec(68.76/2) - 1] = 52.92\ m$$

$$D = 2 * \pi * R * \frac{\Delta}{360}$$

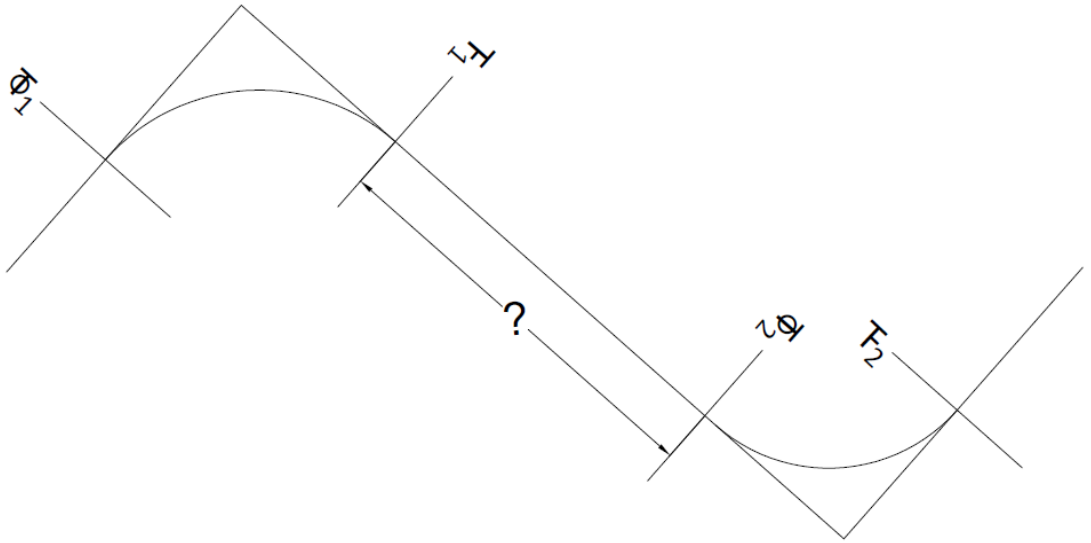
$$t = \mathrm{R} * tg(\frac{\Delta}{2})$$

$$b = R * \left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1\right)$$

b) Calculate the distance between the inverse horizontal curves and check if the distance is enough.

$\overline{F_1\Phi_2} = \Phi_{2,km} - F_{1,km}$ = [1+500] − [1+344] = *156 m* ***≥ 60 m***

it is enough



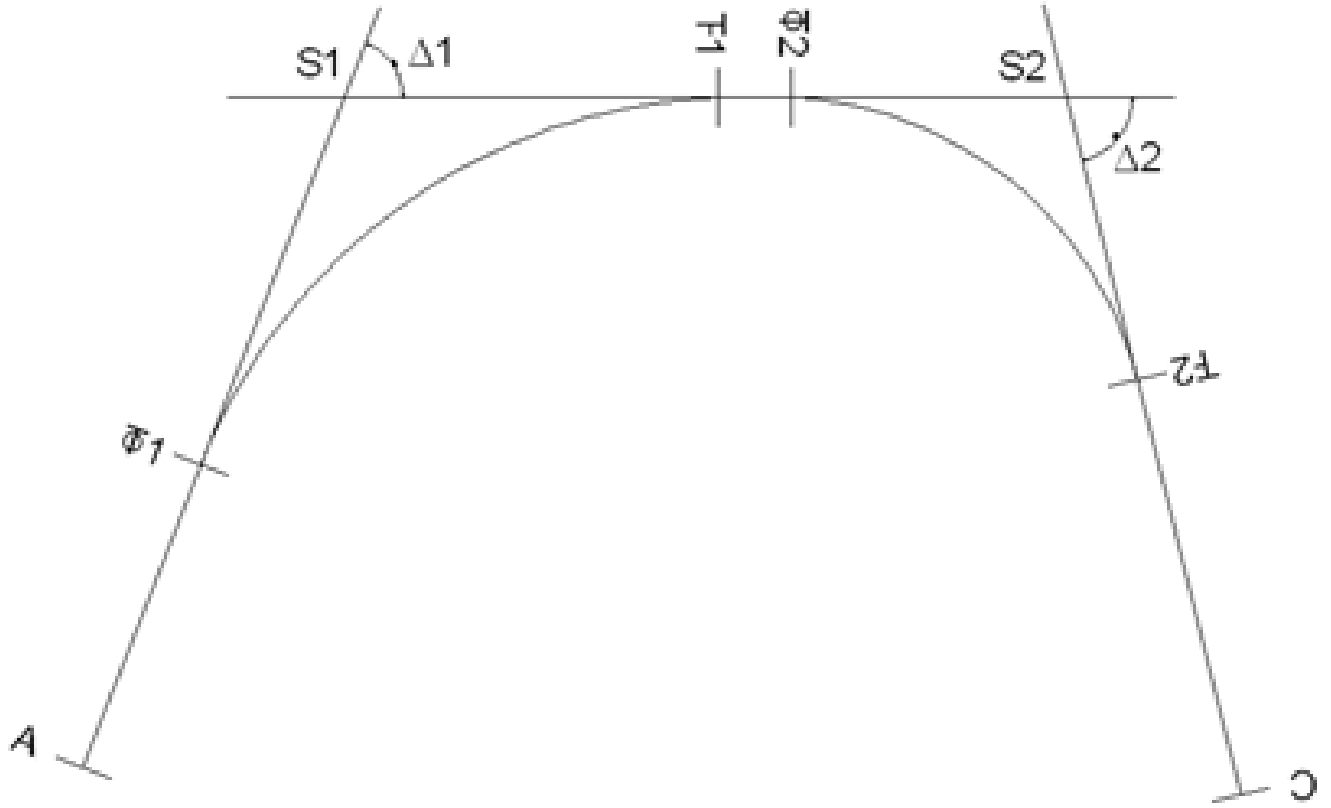
Question – 4

As presented in figure below, a route section is designed with two curves. Radius of the first curve R_1 = 400 m, and curve angle is Δ_1 = 69°; for the second curve, curve angle is Δ_2= 75.26°. According to the existing route geometry, maximum radius is being planned to used for second curve. The distance limit between corresponding curves is 30 meters and the distances between the following points are provided as $\overline{AS_1}$ = $\overline{S_1S_2}$ = $\overline{S_2C}$ = 500 meters.

a) Find missing components of the first curve.

b) Find radius of the second curve.

c) Find total length of the route section.



$$D = 2 * \pi * R * \frac{\Delta}{360}$$

$$t = \mathrm{R} * tg(\frac{\Delta}{2})$$

$$b = R * \left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1 \right)$$

a) Find missing components of the first curve.

$$t_1 = 400 * tan\frac{69}{2} = 274.91\ m$$

$$D_1 = 2 * \pi * 400 * \frac{69}{360} = 481.71\ m$$

$$b_1 = 400 * [\sec(69/2) - 1] = 85.36\ m$$

b) Find radius of the second curve.

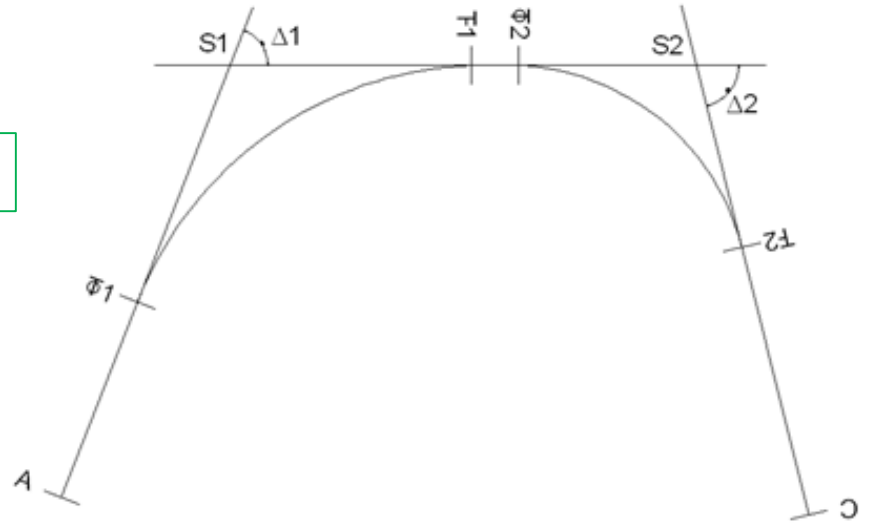
$$t_2 = \overline{S_1 S_2} - t_1 - 30 = 500 - 274.91 - 30 = 195.09\ m$$

$$195.09 = R_2 * tan\frac{75.26}{2} \rightarrow \mathrm{R_2} = 253.06\ \mathrm{m} \rightarrow \mathrm{R_2} = 250\ \mathrm{m}$$

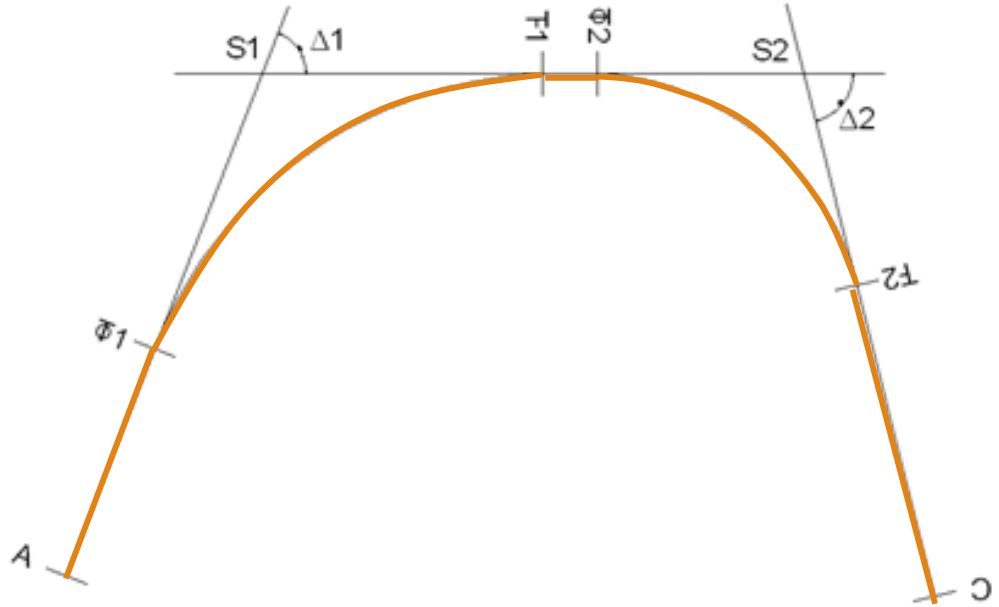
it has to be a multiple of 5

$$t_{2,new} = 250 * tan\frac{75.26}{2} = 192.73\ m$$

$$D_2 = 2 * \pi * 250 * \frac{75.26}{360} = 328.38\ m$$



c) Find total length of the route section.



$$A_{km} = 0 + 000$$

$$\Phi_{1,km} = A_{km} + (\overline{AS_1} - t_1) = [0 + 000] + (500 - 274.91) = 0 + 225^{09}$$

$$T_{1,km} = \Phi_{1,km} + D_1 = [0 + 225^{09}] + 481.71 = 0 + 706^{80}$$

$$\Phi_{2,km} = T_{1,km} + (\overline{S_1S_2} - t_1 - t_{2,new}) = [0 + 706^{80}] + (500 - 274.91 - 192.73) = 0 + 739^{16}$$

$$T_{2,km} = \Phi_{2,km} + D_2 = [0 + 739^{16}] + 328.38 = 1 + 067^{54}$$

$$C_{km} = T_{2,km} + (\overline{CS_2} - t_{2,new}) = [1 + 067^{54}] + (500 - 192.73) = 1 + 374^{81}$$

Total length of the route = 1374.81 m

Question – 5

A two way double lane highway has a project velocity (design speed) of 70 km/h. Some data about this highway and criteria is given below. There is a curve on this road section without transition curve and without ramp. It is required that to perform a constant centerline superelevation on this curve.

a) Find all components of the curve.

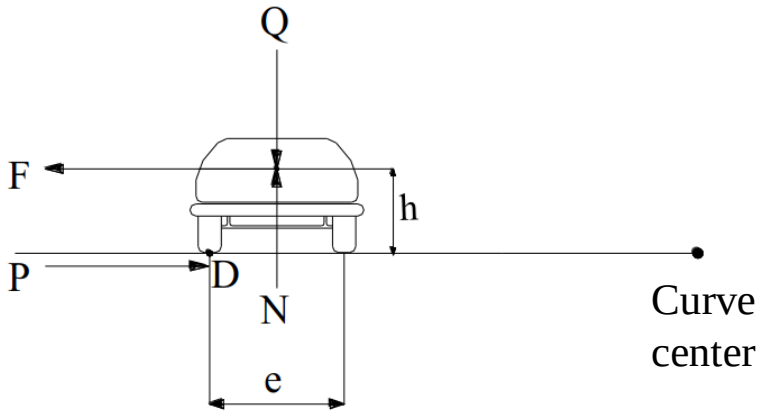
b) By considering that at most half of the centrifugal force can be compensated by the friction, is the curve radius proper for safely passing?

c) Draw the constant centerline superelevation diagram for this curve and find the superelevation heights of the inner and the outer edges of the given cross sections.

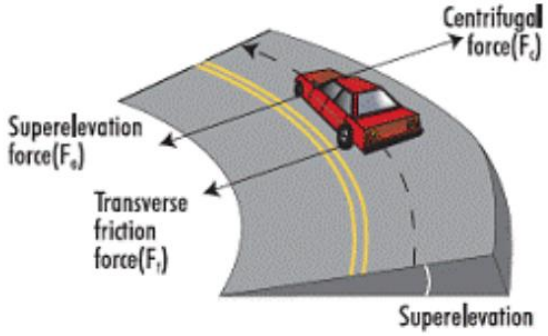
Project Data	Project Criteria
$\Delta = 59.73°$	Platform width = 10 m
$\Phi_{km} = 1 + 000$	Roof slope = %2 ; Max. Superelevation = %8
$T_{km} = 1 + 344$	Theorotical superelevation = $0.786*(V^2/R)$; $L_d = 0.0354*V^3/R$

Cross-section No	Km
50	0 + 990
Φ	1 + 000
51	1 + 010

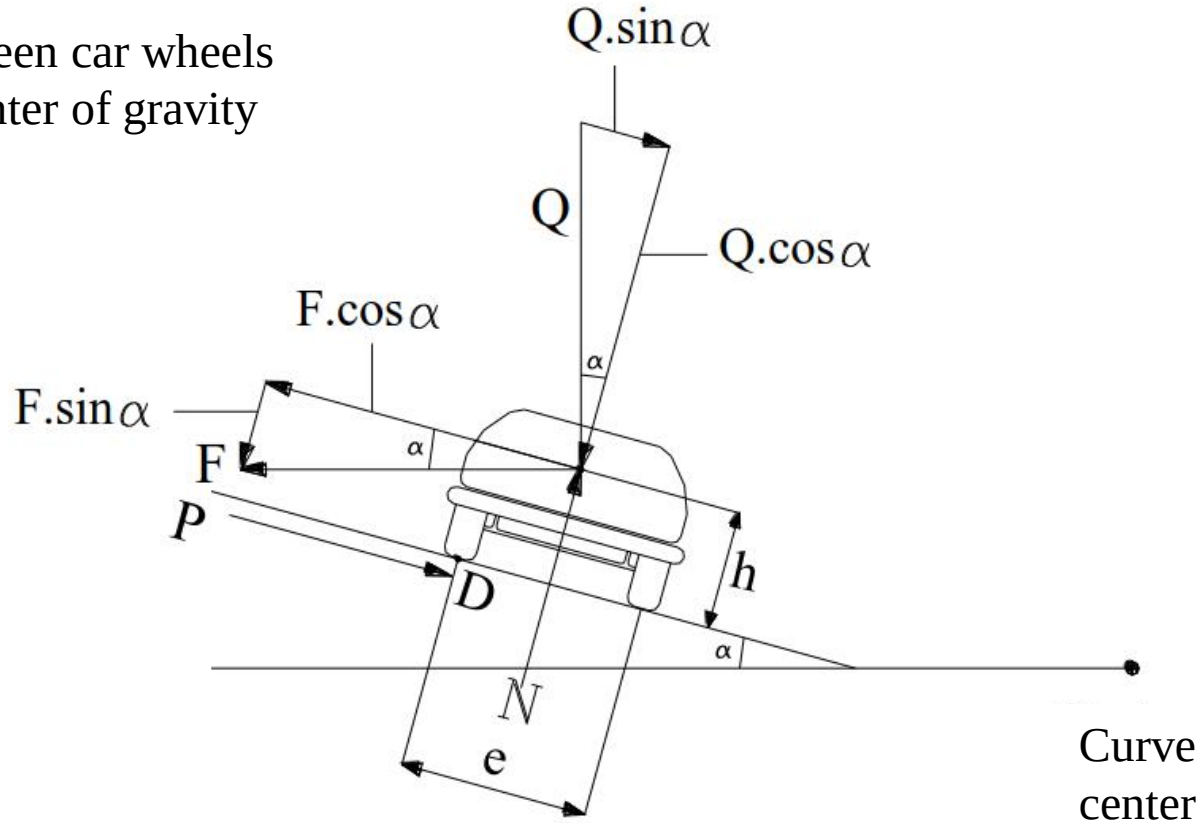
Question – 5



Q: vehicle weight
N: normal force
P: lateral frictional force
F: centrifugal force
e: lateral distance between car wheels
h: distance between center of gravity and road surface







a) Find all components of the curve.

These are calculated in 3. question part a.

$$D = 2 * \pi * R * \frac{\Delta}{360}$$

$$t = \mathrm{R} * tg(\frac{\Delta}{2})$$

$$b = R * \left(\frac{1}{\cos\left(\frac{\Delta}{2}\right)} - 1 \right)$$

$\mathrm{R} = 330\ \mathrm{m}$ $\mathrm{t} = 189.49\ \mathrm{m}$ $\mathrm{b} = 50.53\ \mathrm{m}$

b) By considering that at most half of the centrifugal force can be compensated by the friction, is the curve radius proper for safely passing?

$$q_{practical} = 0.5 * q_{theorotical}$$

$$q_{pra} = 0.5 * 0.786 * \frac{V^2}{R} \rightarrow 8 = 0.5 * 0.786 * \frac{70^2}{R_{min}} \rightarrow R_{min} = 240\ m$$

Curve radius;

$\leq$ ***R = 330 m***

It is proper.

OR

$$q_{pra} = 0.5 * 0.786 * \frac{V^2}{R} = 0.5 * 0.786 * \frac{70^2}{330} = \%5.84$$

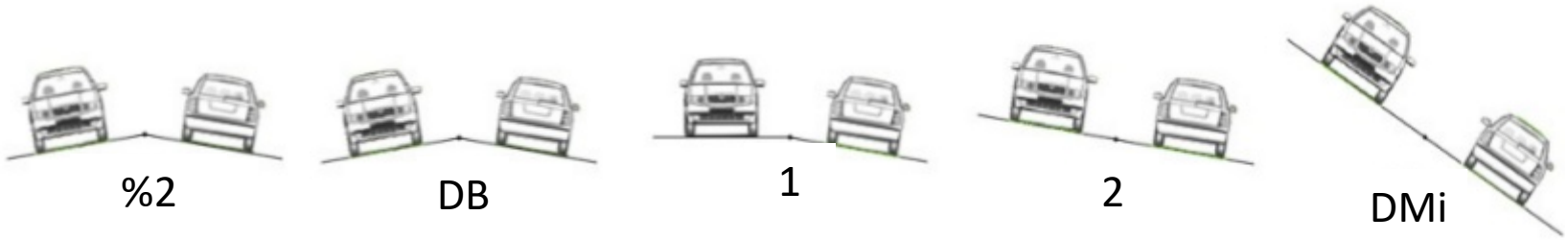
$\leq \%8$ It is proper.

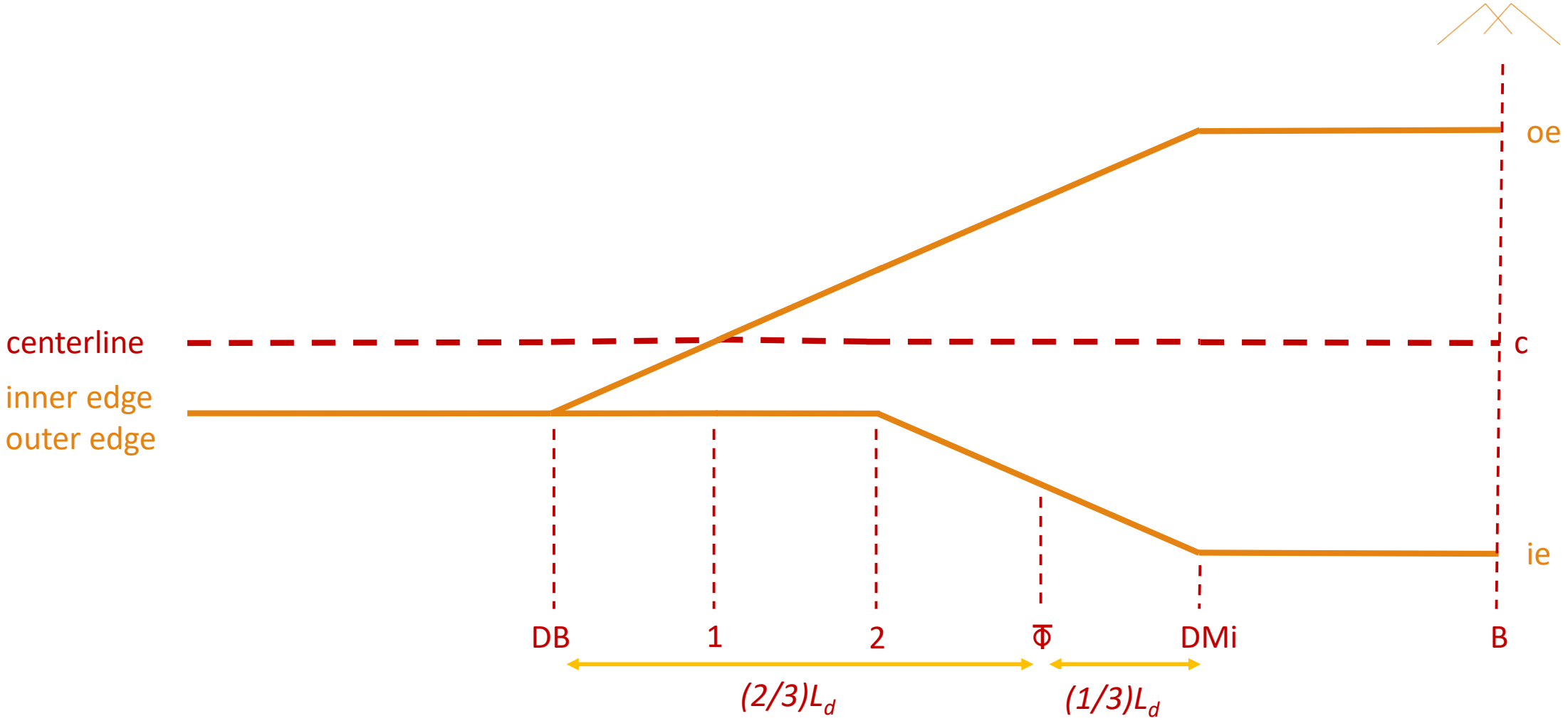
c) Draw the constant centerline superelevation diagram for this curve and find the superelevation heights of the inner and the outer edges of the given cross sections.

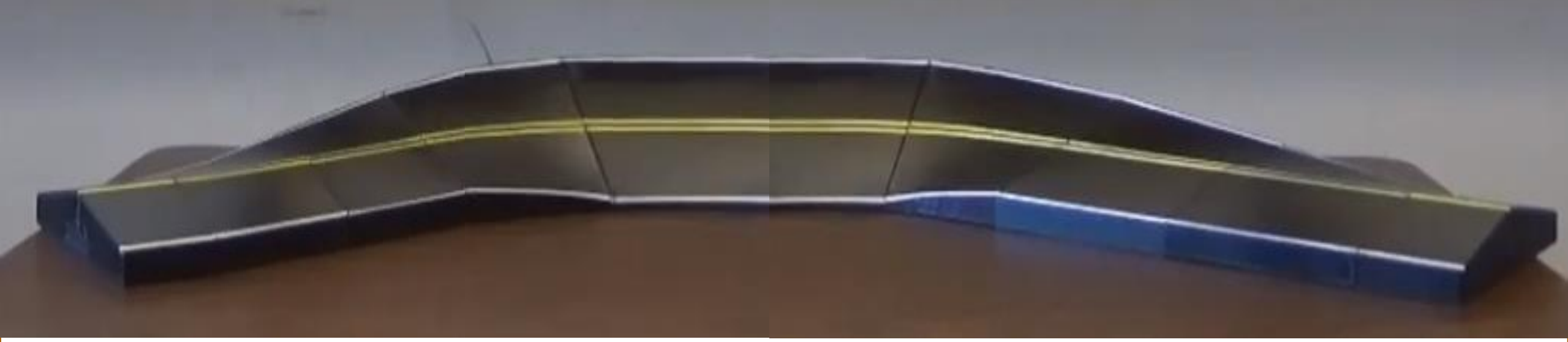
$$q_{pra} = 0.5 * q_{theo}$$

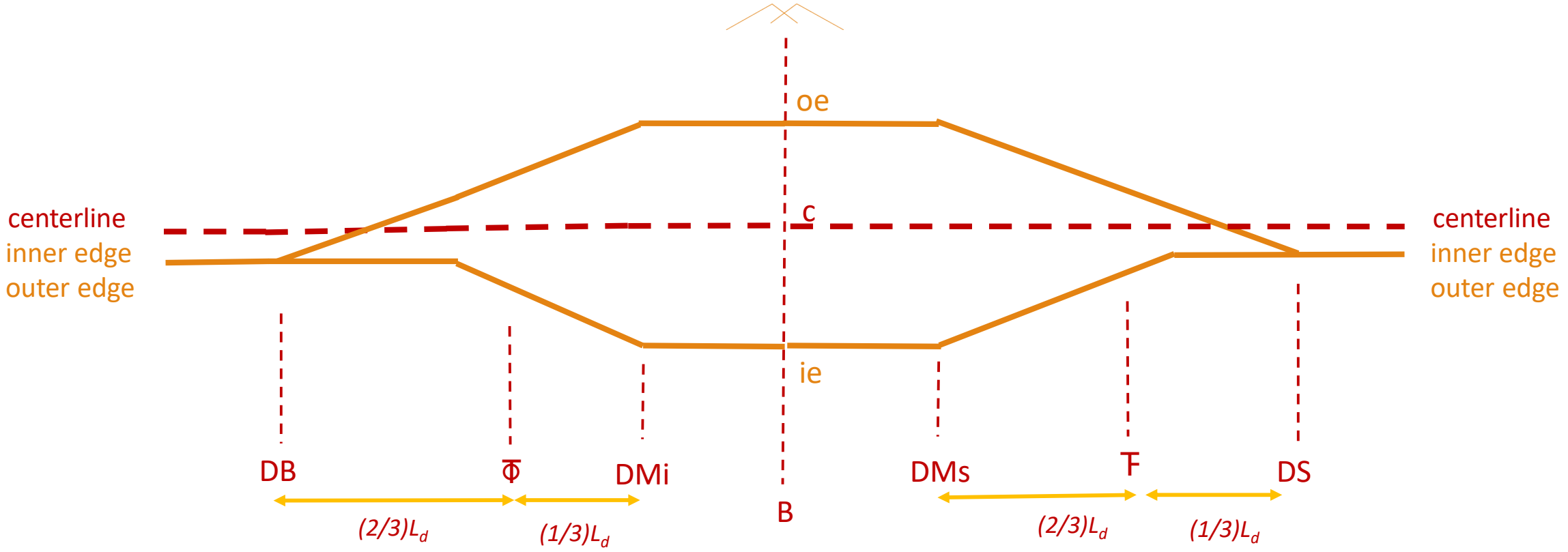
$$q_{pra} = 0.5 * 0.786 * \frac{V^2}{R} = 0.5 * 0.786 * \frac{70^2}{330} = \%5.84 \quad \leq \%8$$

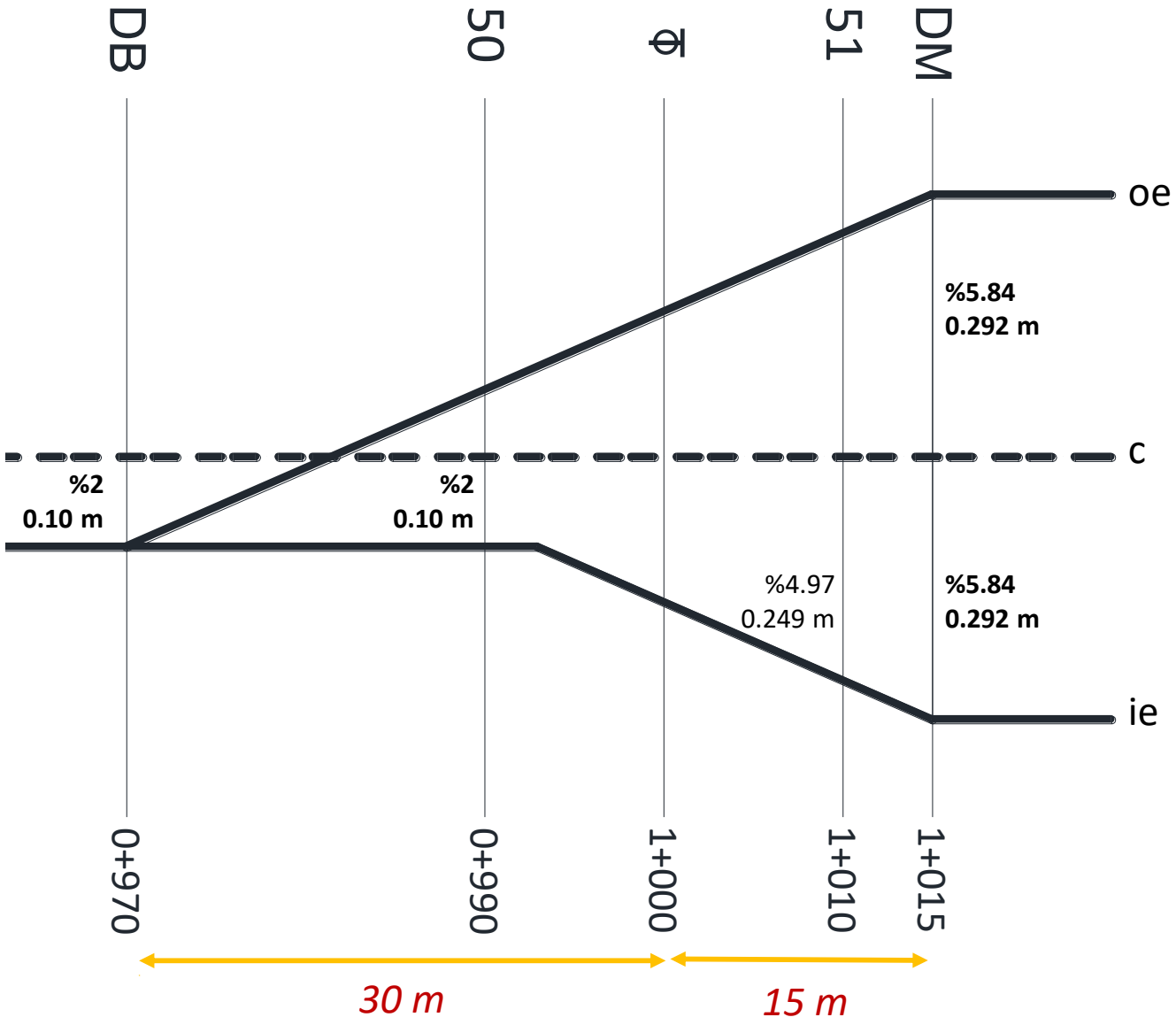
$$L_d = 0.0354 * \frac{V_p^3}{R} = 0.0354 * \frac{70^3}{330} = 36.79\, m \quad \leq \mathbf{45\ m} \longrightarrow \quad So\ L_d\ is\ taken\ as\ L_d = 45\, m$$


%2
DB
1
2
DMi


oe
centerline
c
inner edge
outer edge
ie
DB
1
2
Φ
DMi
B
(2/3)L_d
(1/3)L_d

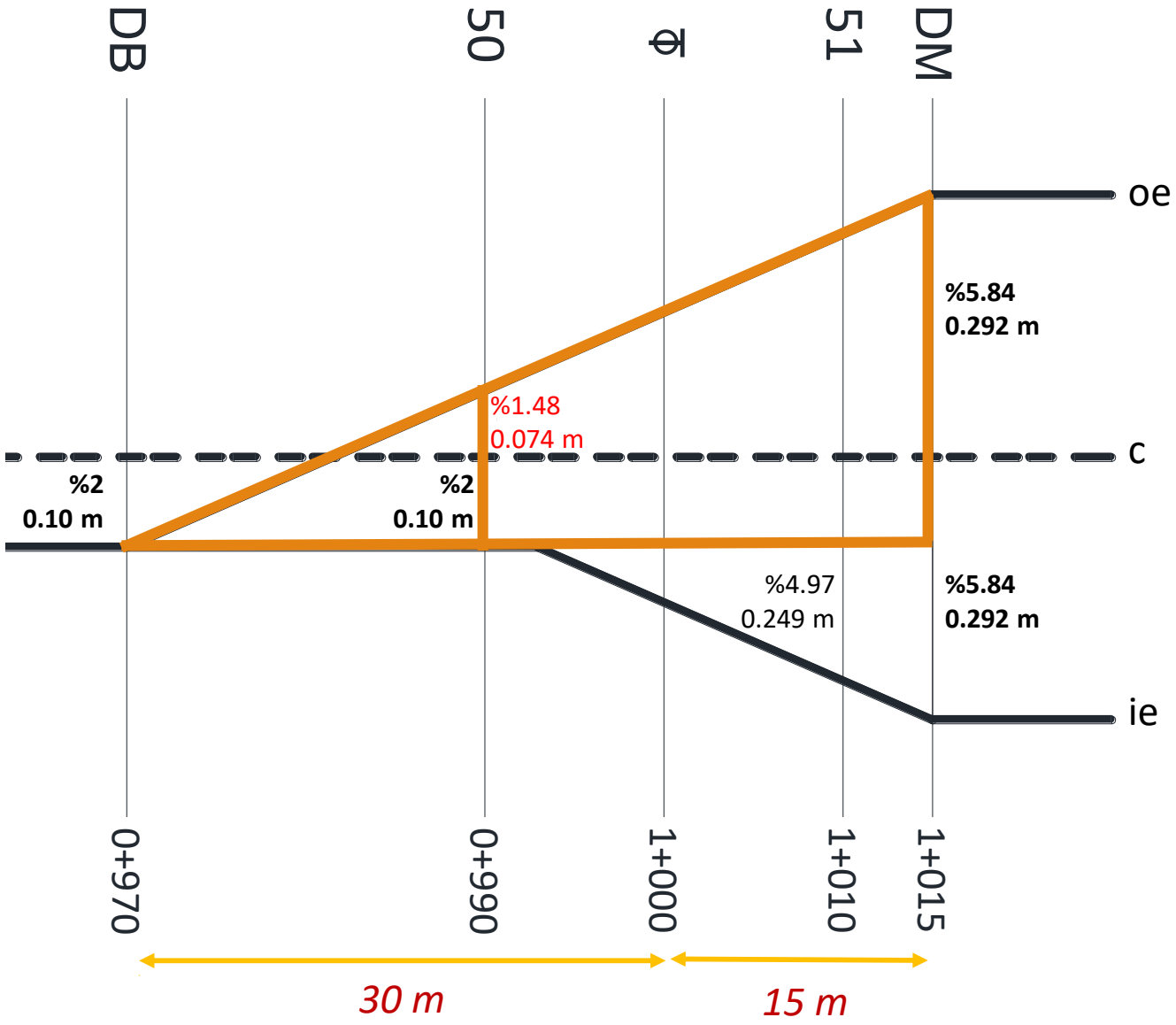



oe
centerline
inner edge
outer edge
c
centerline
inner edge
outer edge
ie
DB
Φ
DMi
B
DMs
F
DS
(2/3)L_d
(1/3)L_d
(2/3)L_d
(1/3)L_d



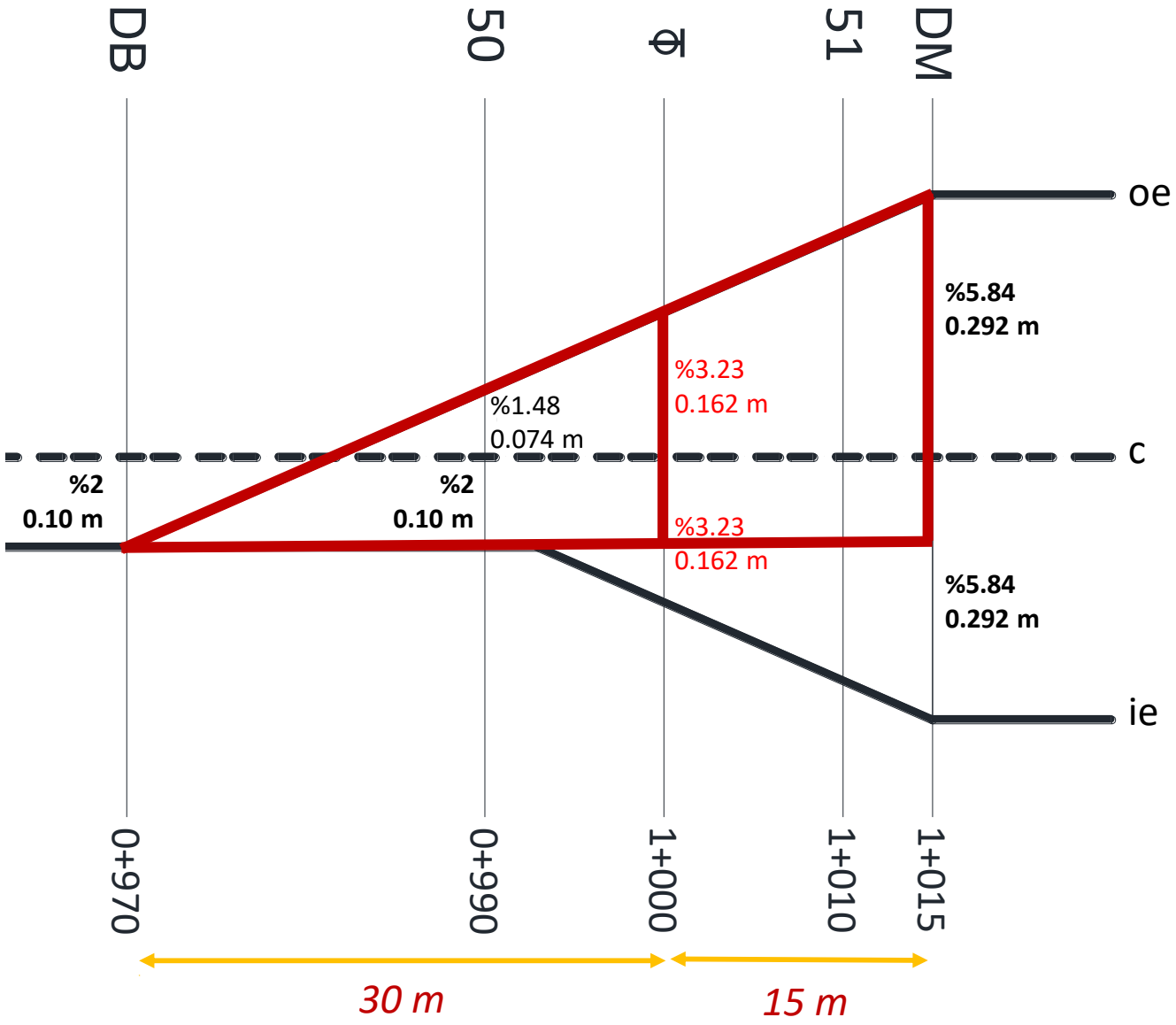
$$h_{\min} = \frac{b}{2} \times q_{\min} = \frac{10}{2} \times 0{,}02 = 0{,}10m$$

$$h_{maks} = \frac{b}{2} \times q_{maks} = \frac{10}{2} \times 0{,}0584 = 0{,}292m$$



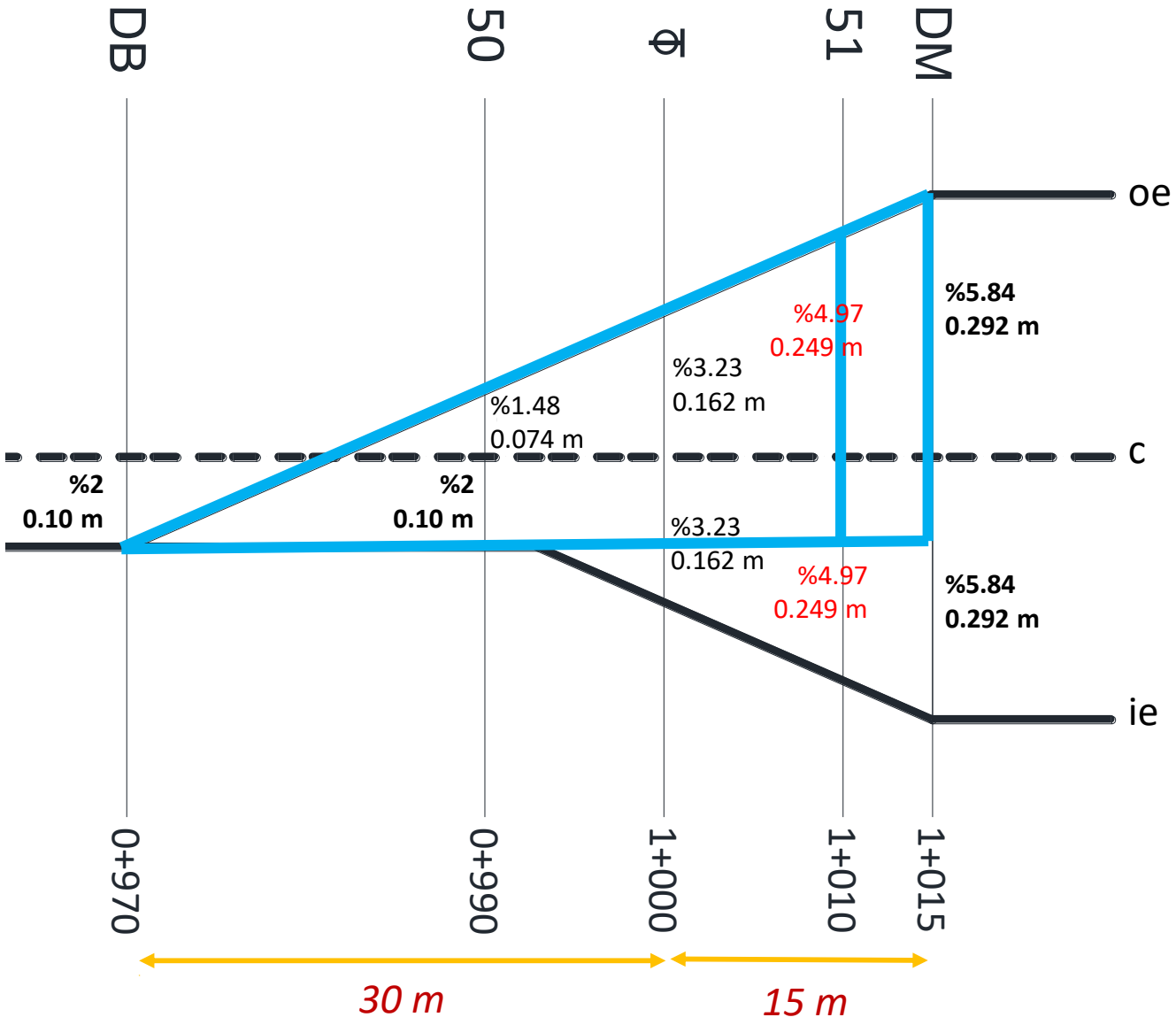
$$\frac{5,84-(-2)}{45}=\frac{x-(-2)}{20}\Rightarrow x=\%1,48$$

$$h_{50}=\frac{10}{2}\times 0,0148=0,074\,\text{m}$$



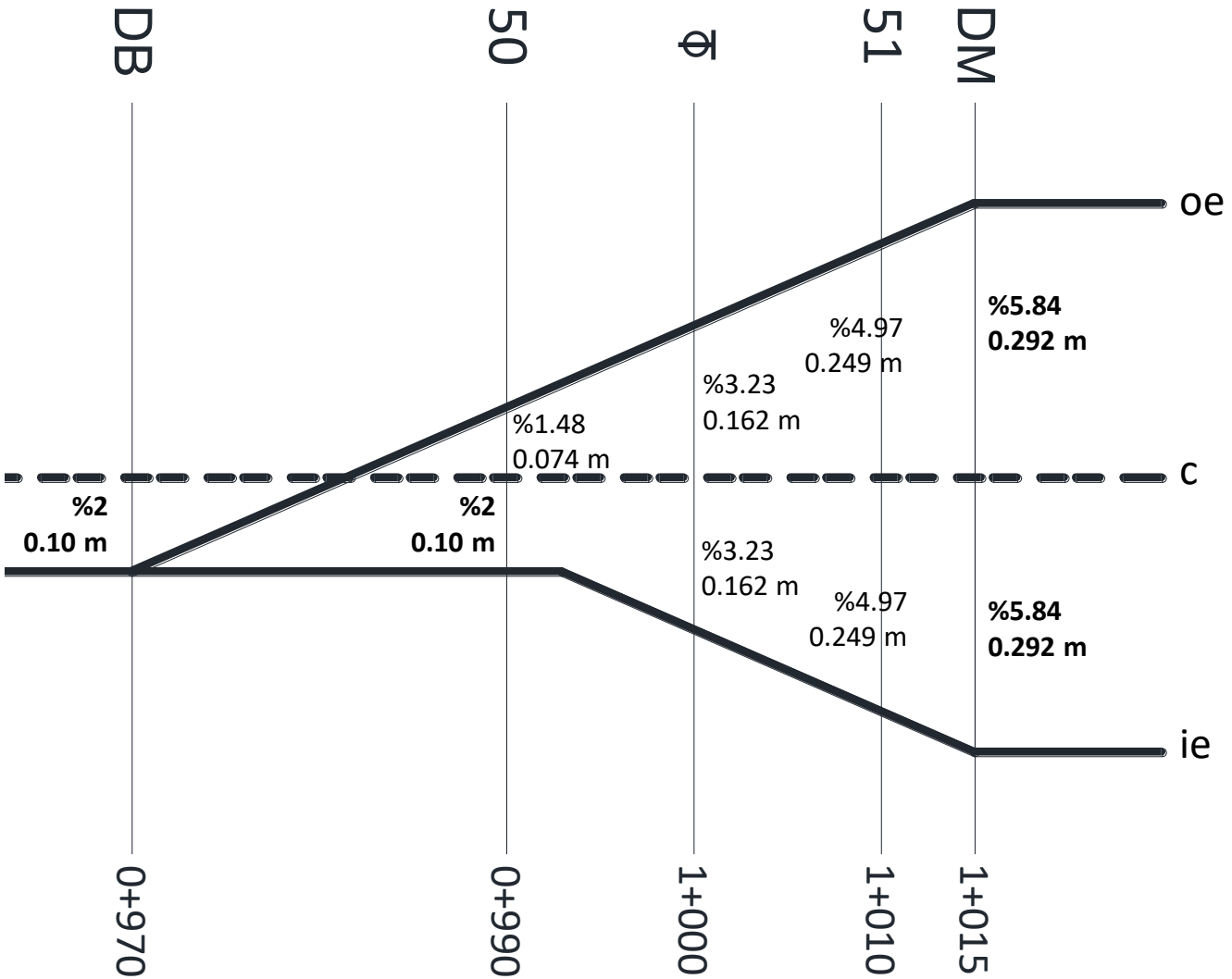
$$\frac{5,84-(-2)}{45}=\frac{x-(-2)}{30}\Rightarrow x=\%3,23$$

$$h_{\Phi}=\frac{b}{2}\times q_{\Phi}=\frac{10}{2}\times 0,0323=0,162\,m$$



$$\frac{5,84-(-2)}{45}=\frac{x-(-2)}{40}\Rightarrow x=\%4,97$$

$$h_{51}=\frac{10}{2}\times 0,0497=0,2485\ \text{m}$$


DB
50
Φ
51
DM
oe
c
ie
%2
0.10 m
%2
0.10 m
%1.48
0.074 m
%3.23
0.162 m
%4.97
0.249 m
%5.84
0.292 m
%3.23
0.162 m
%4.97
0.249 m
%5.84
0.292 m
0+970
0+990
1+000
1+010
1+015